

```
from mlxtend.frequent_patterns import apriori, association_rules
from mlxtend.preprocessing import TransactionEncoder
import pandas as pd
```

```
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```
dataset = [
    ['Milk', 'Bread', 'Butter'],
    ['Beer', 'Bread'],
    ['Milk', 'Bread', 'Beer', 'Butter'],
    ['Bread', 'Butter'],
    ['Milk', 'Butter'],
    ['Beer', 'Chips'],
    ['Milk', 'Bread', 'Butter', 'Chips'],
    ['Bread', 'Milk'],
    ['Milk', 'Chips'],
    ['Bread', 'Butter']
]
```



```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timez
return datetime.datetime.utcnow().replace(tzinfo=utc)
```

```
rules = association_rules(frequent_items, metric="confidence", min_threshold=0.6)
print(frequent_items)
print(rules)
```

```
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	support	itemsets
0	0.3	(Beer)
1	0.7	(Bread)
2	0.6	(Butter)
3	0.3	(Chips)
4	0.6	(Milk)
5	0.5	(Bread, Butter)
6	0.4	(Bread, Milk)
7	0.4	(Milk, Butter)
8	0.3	(Bread, Milk, Butter)

	antecedents	consequents	antecedent support	consequent support	\
0	(Bread)	(Butter)	0.7	0.6	
1	(Butter)	(Bread)	0.6	0.7	
2	(Milk)	(Bread)	0.6	0.7	
3	(Milk)	(Butter)	0.6	0.6	
4	(Butter)	(Milk)	0.6	0.6	
5	(Bread, Milk)	(Butter)	0.4	0.6	
6	(Bread, Butter)	(Milk)	0.5	0.6	
7	(Butter, Milk)	(Bread)	0.4	0.7	

	support	confidence	lift	representativity	leverage	conviction	\
0	0.5	0.714286	1.190476	1.0	0.08	1.4	
1	0.5	0.833333	1.190476	1.0	0.08	1.8	
2	0.4	0.666667	0.952381	1.0	-0.02	0.9	
3	0.4	0.666667	1.111111	1.0	0.04	1.2	
4	0.4	0.666667	1.111111	1.0	0.04	1.2	
5	0.3	0.750000	1.250000	1.0	0.06	1.6	
6	0.3	0.600000	1.000000	1.0	0.00	1.0	
7	0.3	0.750000	1.071429	1.0	0.02	1.2	

	zhangs_metric	jaccard	certainty	kulczynski
0	0.533333	0.625000	0.285714	0.773810
1	0.400000	0.625000	0.444444	0.773810
2	-0.111111	0.444444	-0.111111	0.619048
3	0.250000	0.500000	0.166667	0.666667
4	0.250000	0.500000	0.166667	0.666667
5	0.333333	0.428571	0.375000	0.625000
6	0.000000	0.375000	0.000000	0.550000
7	0.111111	0.375000	0.166667	0.589286

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